

HIGH PERFORMANCE SRAM DESIGN FOR ENERGY EFFICIENT SYSTEMS

S13APROJ2 – PROJECT WORK (PHASE – II)

Submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in Electronics and Communication Engineering

by

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(DEEMED TO BE UNIVERSITY)**

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the Bonafide work of **N UMESH SURYA KIRAN (42130311)**, **VADDI UGANDHAR (42130516)** who carried out project entitled **High Performance SRAM Design for Energy Efficient Systems** under my guidance from September 2025 to April 2026.

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DECLARATION

We **N UMESH SURYA KIRAN (42130311), VADDI UGANDHAR (42130516)** hereby declare that the project entitled **High Performance SRAM Design for Energy Efficient Systems** done by us under the guidance of **Dr. T. RAVI, M.E., Ph.D.**, at **SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI** is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in Electronics and Communication Engineering.

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SIGNATURE OF CANDIDATES

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ABSTRACT

Static Random Access Memory (SRAM) plays a critical role in modern integrated circuits, particularly in applications requiring high-speed and low-power operation such as Internet-of-Things (IoT) devices, portable electronics, and edge computing systems. With the increasing demand for energy-efficient designs, aggressive supply voltage scaling has become a widely adopted technique to reduce power consumption. However, voltage scaling significantly impacts SRAM performance by degrading noise margins, increasing susceptibility to process variations, and reducing reliability during read and write operations. Conventional SRAM designs face a fundamental trade-off between stability, power consumption, and area, making it challenging to achieve optimal performance under low-voltage conditions. In this work, a custom optimized 6T SRAM cell is proposed to address these challenges through intrinsic circuit-level optimization. The design focuses on improving read stability, write ability, and reducing leakage power by carefully optimizing transistor sizing and balancing the strengths of pull-up, pull-down, and access transistors. Unlike conventional approaches that rely on assist techniques or additional circuitry, the proposed design enhances performance within the standard 6T structure, thereby maintaining area efficiency and simplicity. The proposed 6T Architecture cell is designed and simulated using Cadence Virtuoso and evaluated under low-voltage operating conditions. Key performance parameters such as power consumption, read delay, write delay, and static noise margin (SNM) are analyzed and compared with an existing 7T SRAM design. Simulation results demonstrate that the proposed design achieves significant reduction in power consumption and leakage current while maintaining comparable read performance and improved write speed. The optimized design also exhibits sufficient noise margin, ensuring reliable operation under reduced supply voltage conditions.

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